

John Ragland

📍 Woods Hole, MA

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Education

BS	Auburn University , Electrical Engineering • Graduated Summa Cum Laude	2019
MS	Auburn University , Electrical Engineering • Thesis : Digital Simulation and Recreation of a Vacuum Tube Guitar Amp url ↗ • Advisor : Thaddeus Roppel • Emphasis Digital Signal Processing, Real-time Audio Processing, Physical Modeling	2020
PhD	University of Washington , Electrical Engineering • Thesis : Using coherent ambient sound to probe the ocean url ↗ • Advisor : Shima Abadi • Emphasis : Acoustic Oceanography: Ambient noise interferometry and ocean acoustic tomography	2024

Experience

- **Postdoctoral Fellow**, Woods Hole Oceanographic Institution, Woods Hole, MA (2025 - present)
- **Postdoctoral Scholar**, University of Washington, Seattle, WA (2024 - 2025)
- **Graduate Researcher**, University of Washington, Seattle, WA (2020 - 2024)
- **Graduate Summer Researcher**, Applied Research in Acoustics, Seattle, WA (2022)

Peer Reviewed Publications

- (in prep) Hoekstra, **Ragland**, Colosi, Bonnel, Park, Dzieciuch, Alford, Bellerjeau, Voet (2026) *Long range, low frequency source localization across the Gulf Stream front using multiple bottom/surface bounce paths observed on a large aperture vertical array*, Journal of the Acoustical Society of America
- (in prep) **Ragland**, Colosi, Dzieciuch (2026) *Simultaneous ocean acoustic inversion and source localization for transmission across the Gulf Stream*, Journal of the Acoustical Society of America
- (in review) **Ragland**, Colosi, Hoekstra, Dzieciuch, Alford, Bellerjeau (2026) *Analysis of acoustic fluctuations for 150-km, low frequency transmissions across the Gulf Stream in the vicinity of the New England Seamount chain*, Journal of the Acoustical Society of America, doi: [10.22541/essoar.15006016/v1](https://doi.org/10.22541/essoar.15006016/v1) [↗](#)
- Schönauf, Hiron, **Ragland**, Raja, Skitka, Solano, Xu, Arbic, Buijsman, Chassignet, Coelho, Helber, Shriver, Summers, Verlinden, Wallcraft (2025) *How Do Tides Affect Underwater Acoustic Propagation: A collaborative approach to improve internal wave modelling at basin to global scales*, Oceanography Magazine 38, doi: [10.5670/oceanog.2025.308](https://doi.org/10.5670/oceanog.2025.308) [↗](#)
- **Ragland**, Abadi (2025) *Characterizing wind-dependent low-frequency ambient sound with ocean observatories initiative hydrophones*, JASA Express Letters 5, 110801, doi: [10.1121/10.0039811](https://doi.org/10.1121/10.0039811) [↗](#)
- **Ragland**, Abadi, Durofchalk, Dall'Osto, Gemba (2025) *Receptions of Kauai Beacon transmissions by ocean observatories initiative hydrophones*, The Journal of the Acoustical Society of America 158, 1113–1124, doi: [10.1121/10.0038971](https://doi.org/10.1121/10.0038971) [↗](#)
- **Ragland**, Abadi, Sabra (2024) *Using Ocean Ambient Sound to Measure Local Integrated Deep Ocean Temperature*, Geophysical Research Letters 51, e2024GL108943, doi: [10.1029/2024GL108943](https://doi.org/10.1029/2024GL108943) [↗](#)
- **Ragland**, Abadi (2022) *Exploring surface source contributions to ocean ambient noise interferometry with airgun shots*, The Journal of the Acoustical Society of America 152, 3069–3077, doi: [10.1121/10.0015231](https://doi.org/10.1121/10.0015231) [↗](#)
- **Ragland**, Schwock, Munson, Abadi (2022) *An overview of ambient sound using Ocean Observatories Initiative hydrophones*, The Journal of the Acoustical Society of America 151, 2085–2100, doi: [10.1121/10.0009836](https://doi.org/10.1121/10.0009836) [↗](#)
- **Ragland**, Abadi, Sabra (2022) *Long-term noise interferometry analysis in the northeast Pacific Ocean*, The Journal of the Acoustical Society of America 151, 194–204, doi: [10.1121/10.0009232](https://doi.org/10.1121/10.0009232) [↗](#)

Grants and Fellowships

(in review) Collaborative Research: Global ocean model internal wave validation with acoustic propagation

NSF

- *Lead PI:* **John Ragland** Woods Hole Oceanographic Institution
- *Co-PI:* John Colosi; Woods Hole Oceanographic Institution
- *Co-PI:* Brian Arbic; University of Michigan
- *Co-PI:* Yadidya Badarvada; Florida State University

Analysis and modeling for Task Force Ocean New England Seamounts Experiments

ONR

2025 – 2027

- *PI:* John Colosi; Woods Hole Oceanographic Institution
- *Co-PI:* **John Ragland** Woods Hole Oceanographic Institution

Analysis of Kauai Beacon Receptions with OOI hydrophones

ONR

2024 – 2027

- *PI:* Shima Abadi; University of Washington
- *Collaborator:* **John Ragland** helped develop project idea and collaborated on the research statement. Grant supported postdoctoral appointment at UW

Community Service

- **Associate Editor** - [Proceedings of Meetings on Acoustics](#) [↗](#), AIP, *Acoustic Oceanography*
- **AOPE department representative** - [WHOI postdoctoral association](#) [↗](#), elected position

Invited Talks

- Advanced Undersea Systems and Technology Group, Lincoln Laboratory, Lexington MA (2026)
- AOPE department, Woods Hole Oceanographic Institution, Woods Hole MA (2026)
- MG&G Group, University of Washington, Seattle WA (2024)
- Navy Research Laboratory, Ocean Sciences Division, Stennis MS (2023)
- MG&G Group, University of Washington, Seattle WA (2023)
- Applied Research Laboratory - UW, Seattle WA (2022)

Awards

- **ONR Postdoctoral fellowship in Acoustics** (2025) - Office of Naval Research: Fellowship awarded to exceptional early-career researchers who wish to continue research in ocean acoustics or the related disciplines of undersea signal processing, marine structural acoustics and transducer materials science
- **eScience postdoctoral fellowship** (Sept 2024) - University of Washington, eScience Institute: Fellowship awarded to interdisciplinary researchers who are actively involved in developing and/or utilizing advanced data science tools and techniques in their research at the UW
- **The Daoma and Murray Strasberg Memorial Scholarship** (May 2023) - Acoustical Society of America: Awarded to exceptional graduate students in ocean acoustics with research relevant to naval applications to ocean acoustics
- **ASA best student paper** (Dec 2022) - Acoustical Society of America: Second place at the ASA Nashville in underwater acoustics technical committee

Media Coverage

- [Listening to the ocean to measure the impact of climate change](#) [↗](#)
- OOI Science Highlights: [An Overview of Ambient Sound Using OOI Hydrophones](#) [↗](#)

Cruise Experience

- RC0090, 2022, 2 days - deployed mooring with two hydrophones that was recovered one week later. The goal of this deployment was to acoustically measure methane seeps in the Puget Sound.
- RR2411, 2024, 21 days - joint operation to measure deep scattering layer, and low-frequency acoustic propagation around seamounts in the North Atlantic.

- AR90, 2025, 21 days - recovered NESMA acoustic moorings in North Atlantic

Open Source Software Contributions

- **OOIPy** - python package for accessing OOI hydrophone data [GitHub](#) [PyPI](#) [DOI](#)
- **xrsignal** - python package that ports functionality from scipy.signal to xarray and is compatible with distributed computing [GitHub](#) [PyPI](#)
- **pygenray** - native python ray tracing code [GitHub](#) [PyPI](#) [DOI](#)

Conference Presentations

- *A Bayesian approach to long-range source localization in the complex propagation environment across the Gulf Stream front* ([10.1121/10.0045262](#)) - Hoekstra, Bonnel, Colosi, Alford, Bellerjeau, Dzieciuch, Park, **Ragland**, Voet (2026) - 190th Meeting of the Acoustical Society of America
- *First steps toward simultaneous Bayesian environmental inversion and source localization for acoustic transmissions across the Gulf Stream front* ([10.1121/10.0045264](#)) - **Ragland**, Colosi, Hoekstra, Dzieciuch, Park, Alford, Bellerjeau (2026) - 190th Meeting of the Acoustical Society of America
- *Observations of sound speed structure on and adjacent to, a seamount in the Gulf Stream extension region of the North Atlantic Ocean* - Colosi, **Ragland**, Torres, Devine, Alford, Voet, Bellerjeau, Miller (2026) - American Geophysical Union, Ocean Sciences Meeting
- *Long-range acoustic fluctuations across the gulf stream front* - **Ragland**, Colosi, Dzieciuch, Park (2026) - American Geophysical Union, Ocean Sciences Meeting
- *Long range, low frequency source localization across the Gulf Stream front using multiple bottom/surface bounce paths observed on a large aperture vertical array: A Bayesian approach* ([10.1121/10.0040221](#)) - Hoekstra, Bonnel, Colosi, Alford, Bellerjeau, Dzieciuch, Park, **Ragland**, Voet (2025) - 189th Meeting of the Acoustical Society of America, Honolulu, HI
- *Observations of acoustic intensity and phase fluctuations for low-frequency acoustic transmissions through the Gulf Stream* ([10.1121/10.0040674](#)) - **Ragland**, Colosi, Park, Dzieciuch (2025) - 189th Meeting of the Acoustical Society of America, Honolulu, HI
- *Comparing Kauai Beacon receptions to simulated acoustic propagation* ([10.1121/10.0037361](#)) - **Ragland**, Durofchalk, Dall'Osto, Abadi, Gemba (2025) - 188th Meeting of the Acoustical Society of America, New Orleans
- *Analysis of very low frequency wind driven noise at Ocean Observatories Initiative hydrophones* ([10.1121/10.0037493](#)) - **Ragland**, Phan, Abadi (2025) - 188th Meeting of the Acoustical Society of America, New Orleans
- *Kauai Beacon receptions and analysis with open-access hydrophones in the North Pacific Ocean* ([10.1121/10.0026938](#)) - **Ragland**, Durofchalk, Gemba, Dall'Osto, Abadi (2024) - 186th Meeting of the Acoustical Society of America, Ottawa, CA
- *Towards acoustic observations of ocean basin temperatures using the Kauai beacon and Ocean Observatories Initiative Hydrophones* - **Ragland**, Durofchalk, Abadi, Dall'Osto, Gemba (2024) - Ocean Sciences Meeting 2024, New Orleans, LA
- *Detecting the Kauai source beacon with ocean observatories innitiative hydrophones* ([10.1121/10.0023175](#)) - **Ragland**, Durofchalk, Gemba, Abadi (2023) - 185th Meeting of the Acoustical Society of America, Sydney, Australia
- *Using ocean ambient sound to sense arrival time fluctuations due to temperature* ([10.1121/10.0023334](#)) - **Ragland**, Abadi (2023) - 185th Meeting of the Acoustical Society of America, Sydney, Australia
- *Using distributed acoustic sensing for ocean ambient sound analysis* ([10.1121/10.0018176](#)) - **Ragland**, Douglass, Abadi (2023) - Acoustical Society of America, Chicago, IL
- *Towards estimating water column properties using ambient noise interferometry in the deep ocean* - **Ragland**, Abadi (2023) - Underwater Acoustics Conference and Exposition, Kalamata, Greece
- *Overview of distributed acoustic sensing technology and recently acquired data sets* ([10.1121/10.0018174](#)) - Douglass, **Ragland**, Abadi (2023) - Acoustical Society of America, Chicago, IL
- *Comparing distributed acoustic sensing data with hydrophone recordings* ([10.1121/10.0018175](#)) - Abadi, Douglass, **Ragland** (2023) - Acoustical Society of America, Chicago, IL

- *Long-term ambient noise interferometry in the NE Pacific deep ocean* - **Ragland**, Abadi (2022) - Ocean Sciences Meeting 2022, Online Meeting
- *Overview of ambient noise research and outreach with OOI hydrophones* - **Ragland**, Schwock, Liu, Abadi (2022) - AGU Fall Meeting 2022, Chicago, IL
- *Overview of ocean ambient noise interferometry – Theory and simulation* ([10.1121/10.0016311](#)) - **Ragland**, Abadi (2022) - Acoustical Society of America, Nashville, TN
- *Exploring surface source distributions for ocean ambient noise interferometry with airgun shots* ([10.1121/10.0011063](#)) - **Ragland**, Abadi (2022) - Acoustical Society of America, Denver, CO
- *OOIPy: A Python toolbox for accessing and analyzing data from the Ocean Observatories Initiative* ([10.1121/10.0007845](#)) - Schwock, **Ragland**, Abadi (2021) - Acoustical Society of America, Seattle, WA
- *An overview of ambient sound using OOI hydrophone network* ([10.1121/10.0007594](#)) - **Ragland**, Schwock, Munson, Abadi (2021) - Acoustical Society of America, Seattle, WA
- *Long-term noise interferometry analysis in the northeast Pacific Ocean* ([10.1121/10.0004609](#)) - **Ragland**, Abadi (2021) - Acoustical Society of America, Online Meeting
- *Estimating ocean variables using ambient noise interferometry* ([10.1121/10.0007697](#)) - **Ragland**, Abadi (2021) - Acoustical Society of America, Seattle, WA
- *Ship detection from passive underwater acoustic recordings using machine learning* ([10.1121/10.0007848](#)) - Alvaro, Schwock, **Ragland**, Abadi (2021) - Acoustical Society of America, Seattle, WA